

Biomass Pyrolysis

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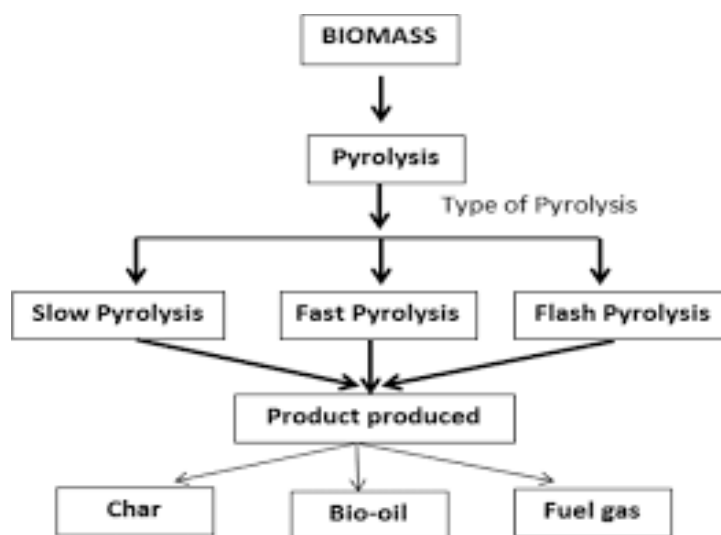
* Definition of Pyrolysis

- pyrolysis is the combination of two Greek words, 'pyro' means 'fire' and 'lysis' means 'separating' i.e. separation by fire or heat.
- it is also called 'thermolysis'.
- Encyclopedia Britannica
 - it is the chemical decomposition of organic (carbon-based) materials through application of heat in the absence of oxygen.
 - in the first step gasification and combustion occur in the absence or near absence of oxygen
 - it is different from combustion which takes place only if sufficient oxygen is present
 - rate of pyrolysis increases with temperature
 - industrial application - temperatures used are often 430°C or higher
 - two well known products - charcoal or biochar created by reacting wood (biomass), coke created by heating coal.
 - also produces condensable liquids (or tar) and non-condensable gases

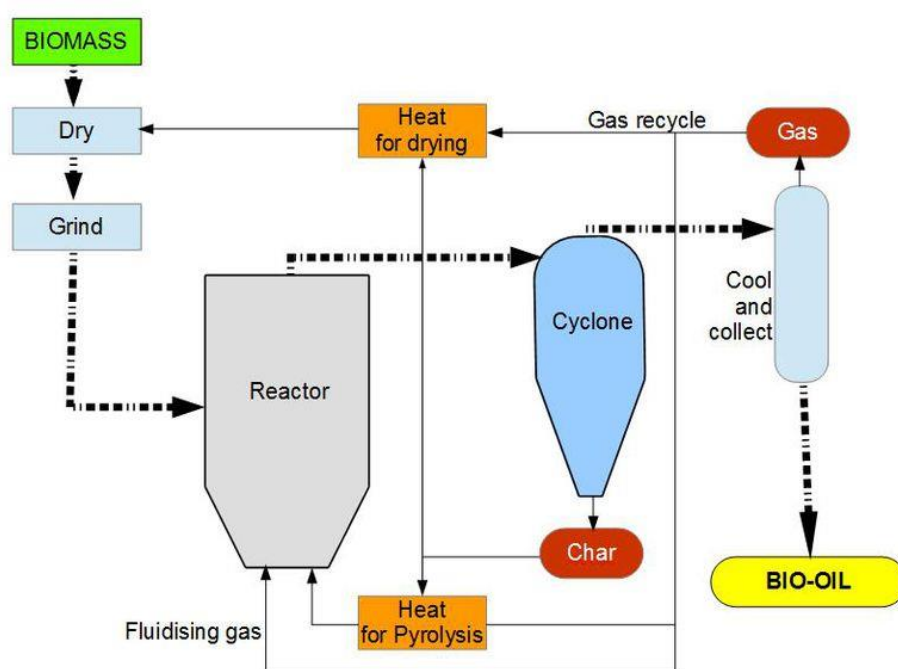
* Biomass Pyrolysis (Figure 1)

- biomass profusely available everywhere is an important renewable energy source
- it is an alternative to partially or replace the fossil fuels and nuclear energy
- 50% of global population depends on biomass for energy
- technological developments in recent years allow the biomass to be used as energy source with low levels of emissions and environmental impacts
- biomass is converted into energy by various technologies
- among all conversion techniques, biomass, pyrolysis offers number of benefits:

Biomass Pyrolysis



(a)



(b)

Figures-1 (a, b): The process of pyrolysis.

Note: Generally, at low temperatures (<450°C) at low heating rate, high yield of bio-char is obtained. However, at high heating and high temperature, gases formed are the primary products. Bio-oils are formed at high heating rates and at intermediate temperature.

- less emissions of pollutants and the flue gas
- gives solid (bio-char), liquid (bio-oil) and gaseous (biogas) products
- all by-products can be re-used
- bio-oil obtained in pyrolysis has advantage over petroleum product due to renewable nature
- bio-oil can be used for further production of many chemicals

- biomass pyrolysis process is carried out in three stages:

- dosing and feeding of the raw materials
- transformation of the organic mass
- separation of the products

- distribution of products is effected by:

- heating rate
- final temperature, and
- pressure
- Residence time

- the products of biomass pyrolysis include bio-char, bio-oil and gases (biogases) containing CH_4 , H_2 , CO , CO_2 depending on temperature, heating rate, residence time process variants with main products are given in the Table-1

Process	Temp (°C)	Heating Rate	Residence Time	Products
Fast	650	Very High	0.5-5 sec.	Bio-oil
Conventional (Slow Pyrolysis)	600	Low	15-30 min	Oil, gas, char
Carbonization	400	Very low	Days	Charcoal
Ultra (Ultrafast Pyrolysis)	1000	Very high	<0.5	Chemicals, Gas
Flash-gas	<650	High	<1 sec	Chemicals, Gas
Flash-liquid	<650	High	<1 sec	Bio-oil
Methanol Pyrolysis	>700	High	<10 sec	Chemicals
Hydro Pyrolysis	<500	High	<10 sec	Bio-oil
Vacuum	400	Medium	2-30 sec	Bio-oil

* Mechanism of Pyrolysis (Figures-2)

- biomass pyrolysis takes place through ^{a number of} ~~series~~ chemical reactions including reactions in parallel as well as in series
- the reactions are dehydration, aromatisation, isomerisation, depolymerization, decarboxylation and char formation
- in general, it is believed that the reaction consists of three stages:
 - the evaporation of free moisture (drying)
 - primary reaction - depolymerization, fragmentation, and formation of char
 - secondary decomposition - depolymerization vapor cracking

✓ ① evaporation of free moisture (see figure-2)

- biomass loses free moisture initially before any physical or chemical conversion
- dehydration starts at 100°C leaving behind amorphous carbon

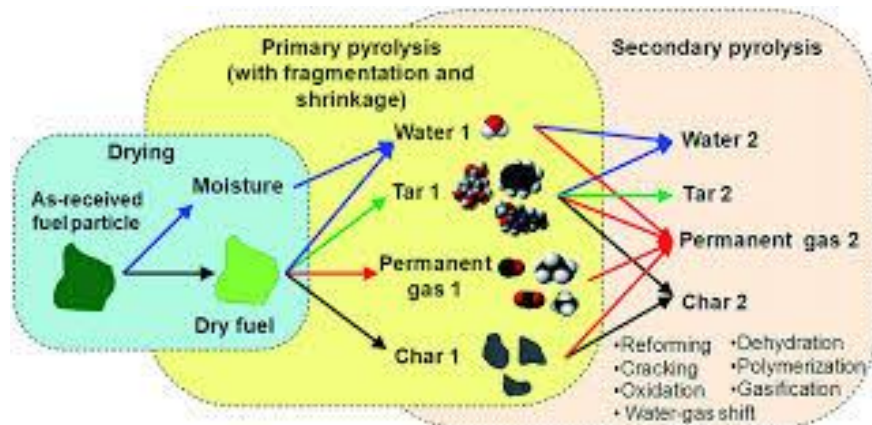
✓ ② primary reactions of biomass pyrolysis

- primary reaction starts after removal of moisture
- chemical bonds are broken in the resulting in the release of some volatiles and rearrangement of reactions
- primary reactions occur which include char formation, depolymerization and fragmentation
- biochar is left after release of condensable and non-condensable gases
- release of volatiles is recovered in the form of liquid fraction.
- reactions occur between $250 - 500^{\circ}\text{C}$

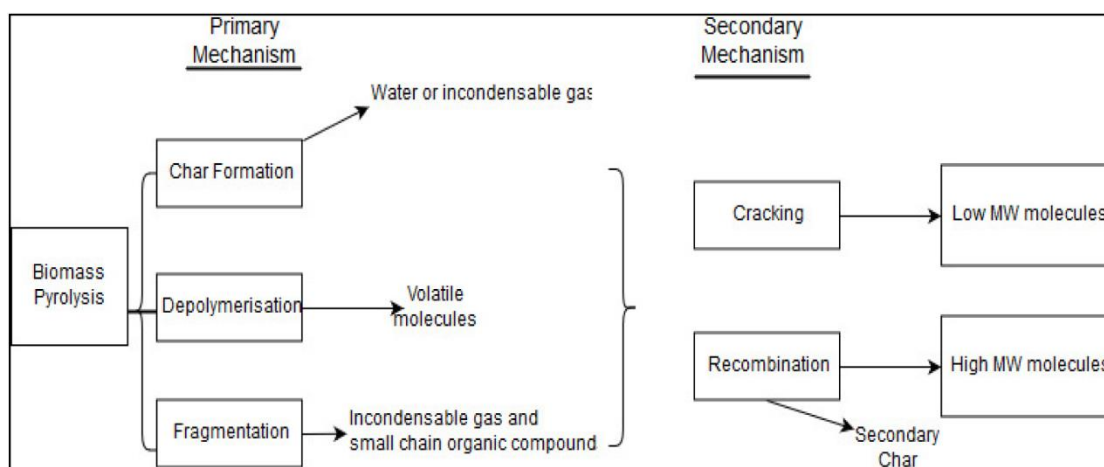
✓ ③ Secondary Degradation

- products generated during all primary reactions are not stable and thus undergo secondary degradation
- reactions under secondary degradation are: generally recombination (repolymerisation) or cracking

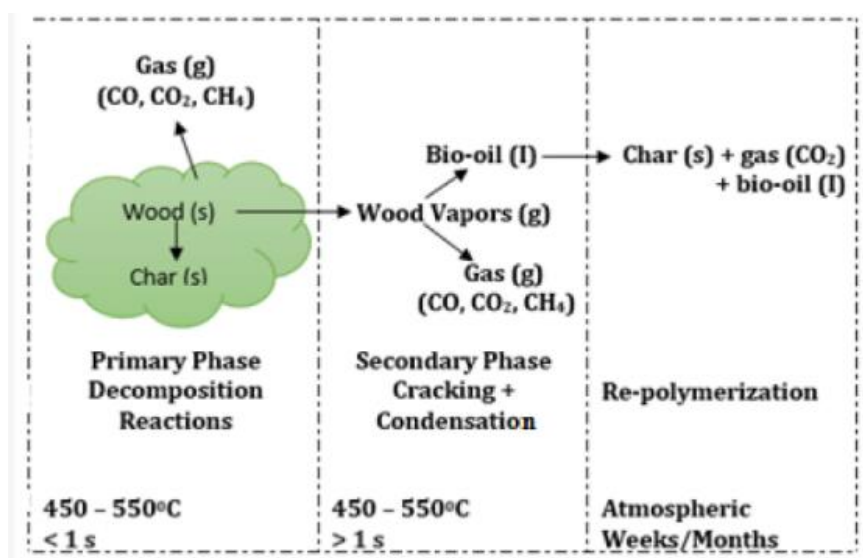
Mechanism of Pyrolysis



(a)



(b)



(c)

Figures-2: Mechanism of pyrolysis.

* Sources of Biomass Feedstock for Pyrolysis

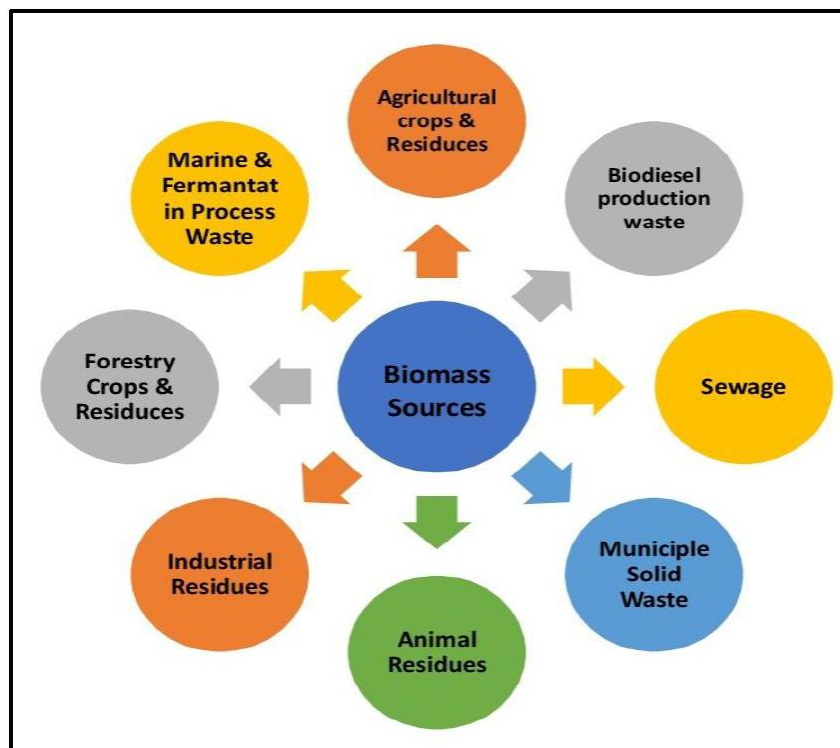
- biomass energy is recognized as the third largest source of energy
- many of the developing countries have been largely occupied by forest and agriculture land where 40-50% of energy is based on biomass
- the sources of biomass include many natural and derived materials such as:
 - agricultural crops and residues
 - forest wood and leaf residues
 - municipal solid wastes
 - forest and mill residues
 - animal residues and sewage
 - aquatic plants, and so on
- the physical and chemical properties of biomass is highly significant for product distribution
- Table-I & Table-II show the properties of some biomass
- figure 3 reflects the available sources of biomass
- highest biochar yield is achieved when the biomass feedstock with high lignin content is pyrolysed under moderate temperature 500°C
- biomass containing high volatile matter generates large amount of syngas (CO , H_2) and bio-oil
- fixed carbon content raises the formation of biochar
- moisture content in biomass effects the heating value and also influences product distribution
- biomass consists of carbon, hydrogen, oxygen and nitrogen.
- in some cases sulfur may also be present
- in some biomass significant amount of inorganic species may also be present
- the chemicals obtained from co products and residues may improve the biomass production chains
- various economic sectors, such as agribusiness, petrochemical, automotive, etc.

Table-II: Physical Characteristics of Biomass

Feedstock	Density (kg/m ³)	Moisture Content (%)	Ash Content (%)	Volatile Matter (%)	Fixed Carbon (%)
Wood	380	20	0.4–1	82	17
Bituminous coal	700	11	8–11	35	45
Wheat straw	18	16	4	59	21
Barley straw	210	30	6	46	18
Pine	124	17	0.03	-	16
Polar	120	16.8	0.007	-	-
Switchgrass	108	13–15	4.5–5.8	-	-

Table-III: Chemical Characteristics of Biomass

Feedstock	Carbon (%)	Hydrogen (%)	Oxygen (%)	Nitrogen (%)	Ash (%)
Wood	51.6	6.3	41.5	0.1	1
Bituminous coal	73.1	5.5	8.7	1.4	9
Wheat straw	48.5	5.5	3.9	0.3	4
Barley straw	45.7	6.1	38.3	0.4	6
Pine	45.7	7	47	0.1	0.03
Polar	48.1	5.30	46.10	0.14	0.007
Switchgrass	44.77	5.79	49.13	0.31	4.30

**Figure-3: Available sources of biomass.**

- note: types of pyrolysis such as rapid, ultrafast and flash pyrolyses have also been discussed in the literature.

* Classification of Pyrolysis

- based on the residence time, rate of heating, particle size and temperature, there are basically three types of pyrolysis:

- slow or conventional
- Fast pyrolysis/Flash Pyrolysis
- Intermediate Pyrolysis

- Table - 4, below demonstrates some operating parameters for three types of pyrolysis process:

TABLE - 4

Process	Time (s)	Rate (K/s)	Size (mm)	Temperature (K)	Oil Yield	Char Yield	Gas Yield
Slow	450-550	0.1-1	5-50	550-950	30	35	35
Fast	0.5-10	10-200	<1	850-1250	50	20	50
Flash	<0.5	>1000	<0.2	1050-1300	75	12	13

- slow pyrolysis or conventional pyrolysis

- the main products are bio-oil, biochar and biogas
- dominated by biochar yield (i.e. the main product)
- temperature is kept as low as 400°C or above 400°C
- operates under high residence time between 5-30 min
- leads to secondary reaction of products in vapor phase with subsequent formation some bio-oil and bio-char

- fast pyrolysis

- occurs with the use of short residence time $\sim 0.5-10$ seconds
- operates at temperatures between $577-977^{\circ}\text{C}$
- very high heating rates are the main features
- depending on the feedstock of biomass, gaseous, liquid and solid products range between 10-20, 60-75 and 15-75% respectively

- intermediate pyrolysis process

- moderate temperature around 500°C is suitable for the process
- the operating condition offers a wide range of variation
- mostly conducted in fixed bed batch reactors for residence time varying from $\sim 10-20$ seconds.

* Pyrolysis Products and their Applications

- pyrolysis of biomass results in three primary products:
 - char or bio-char (solid product)
 - bio-oil (the condensed liquid product)
 - permanent gases or biogas (syngas)
- biomass pyrolysis products yield may be improved
 - biochar yield improvement by applying less temp. keeping lower heating rate
 - bio-oil yield is enhanced by lower temperature but higher heating rate
 - gaseous products are improved by higher temperature at lower heating rate

→ Biochar

- production of biochar is an emerging bioresource technology for improvement of "food security" and mitigation of "climatic change"
- potential benefits of biochar as soil enrichment may be highlighted as: addressing issues such as:
 - i. waste management
 - ii. biorefinery production
 - iii. increased soil fertility through alteration of soil pH
 - iv. retention of nutrients through cation adsorption
 - v. reduction of emission of nitrous oxide (N_2O).
- as a promising modifier to soil, biochar has attracted the attention of policymakers in USA, Japan, Europe and in some developing countries
- concised benefits of biochar for agriculture sector is summarized in Table-5:

→ Bio-oil

- it is the main product of biomass pyrolysis process
- there is a very current demand to convert biomass into bio-oil in order to use in ships, trains and aeroplanes
- reactor designs are the primary target of researchers to obtain a better quality bio-oil

- Reduced nitrogen leaching in groundwater
- Possible reduced emissions of nitrous oxide
- Increased cation exchange capacity resulting in better soil fertility
- Moderation of soil acidity
- Greater water retention
- Increase in the number of beneficial soil microbes

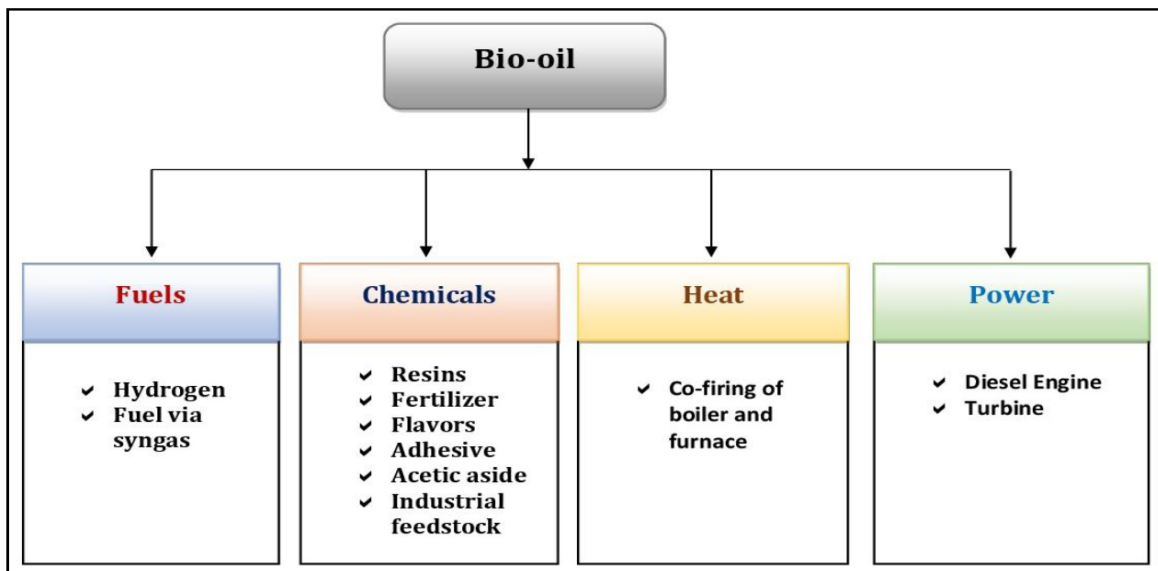
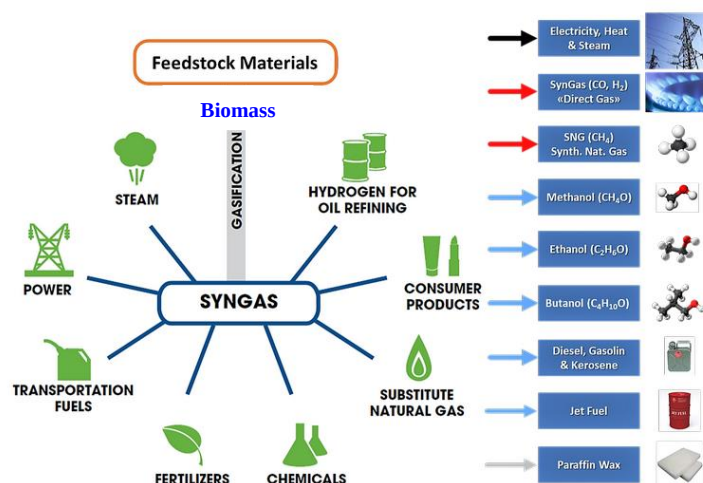


Figure-4: Various applications of pyrolysis bio-oil.

– Biogas (Syngas)

- around 10–35% of biogas is produced in slow pyrolysis process
- it can be sued as an **alternative renewable source of fuel in industrial combustion processes, as well as for internal combustion (IC) engines**
- can be used as fuel in converted commercial petrol and diesel engines
- biogas yield is highly influenced by the pyrolysis temperature, and it is possible to achieve ahigher yield in flash pyrolysis with high temperatures
- higher temperatures favour tar decomposition and the thermal cracking of tar to increase the proportion of biogas
- biogas mainly consists of hydrogen (H₂) and carbon monoxide (CO), it is also known as syngas.**
- it may also contain a **small volume of nitrogen (N₂), water, carbon dioxide (CO₂), hydrocarbons such as C₂H₄, CH₄, C₂H₆, ash,tar, and so on**
- which depend on biomass feedstock and pyrolysis conditions

Broad applications demonstrated in the Figure-5, below:



* Pyrolyzers used in the Pyrolysis Process

- pyrolyzers (reactors) are the heart of biomass pyrolysis process
- it is the core of the pyrolysis technology where lot of attention has been paid by research community for advancement
- currently, many types of reactors have been designed mostly with the aim to maximize the product yield of bio-oil
- lately, many types of pyrolyzers have been developed, the main being the fluidized bed reactors (bubbling & circulating)
- the general classification of pyrolyzers are batch or continuous
- the classification may be summarized as:

- fixed bed reactor
- fluidized bed reactor
- ablative reactor
- vacuum pyrolysis reactor
- rotating cone reactor
- auger pyrolysis reactor

- fixed bed reactor

- it is a simple, reliable, and proven system customarily used for production of biochar (shown in Figure-6)
- consists of a gas cooling and cleaning system
- generally operates with high carbon preservation, low gas velocity, and low residence time

Residence time

- the basic units are drying, granulation, heating and cooling
- the system operates between the temperatures are 450 to 750°C.
- heating rate fluctuates between 5 and 100°C/min
- the temperature ensures that the variables such as temperature program, heating rates, and residence time are within the limit.

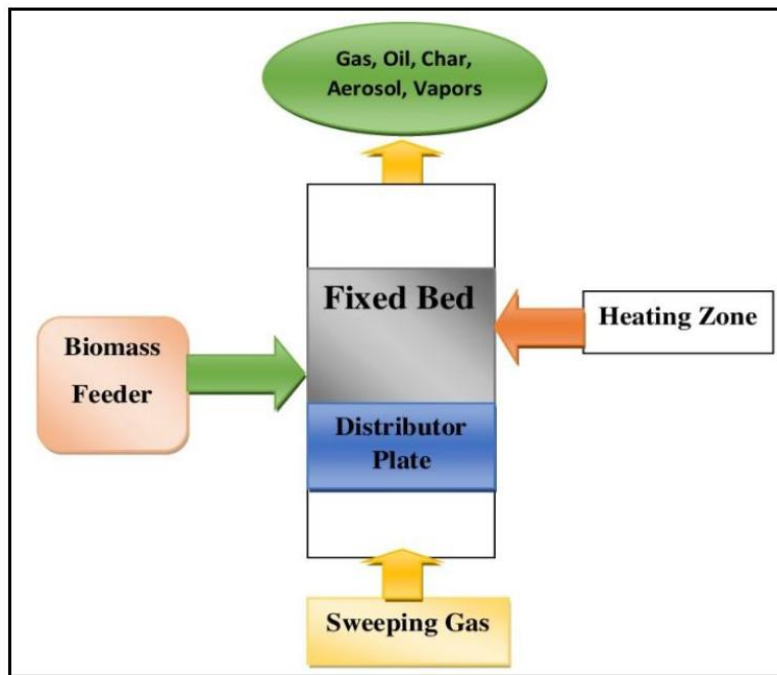
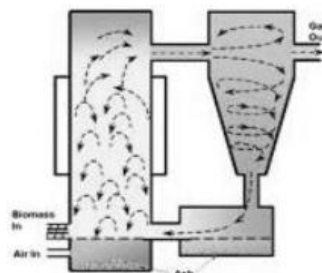


Figure 6. Fixed bed reactor.

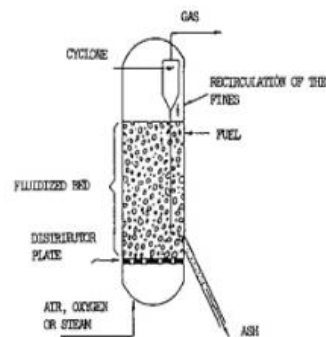
Fluidized Bed Reactor

- fluidized bed reactors (bubbling and circulating) have a well-known technology
- they have a series of industrial applications
- they are used in many projects in order to maximize the liquid product (bio-oil)
- as biomass has a very low density, it is common in fluidized bed reactors to use an inert element, usually sand, to give fluid dynamic stability to the process and help biomass heating
- the fluidised bed reactor comprises a fluid–solid blend that shows similar properties to the fluid
- fluidised bed reactors seem to be widespread and popular because they offer rapid reaction and heat transfer, a wide and high shallow area of contact between the fluid and solid,
- different types of fluidized bed reactors are available which include bubbling fluidized bed reactors and circulating fluidized bed reactors (figures-7)

- Fluidised Bed Gasifier
 - Bubbling fluidised bed
 - Circulating fluidised bed



Circulating fluidised bed



Bubbling fluidised bed

–Ablative Reactor (Figure-8)

- ablative pyrolysis is primarily dissimilar from fluid bed procedures in the absence of a fluidizing gas
- material connected to the wall fundamentally melts, and the residual oil evaporates as pyrolysis vapours.
- the ablative pyrolysis reactors have good heat transfer with high heating rates and a relatively small contact surface.
- they also have high energy and cost efficiency
- as no heating and cooling of fluidizing gases is required, furthermore, they tolerate fixing of condensation units with a small volume in requiring less space at lower costs.
- high pressures are needed for this type of process, and temperatures are usually under 600 °C; there is no ceiling to the particle size parameter for this type of reaction, but this type of pyrolysis reaction requires very high heat at the reactor wall.
- this type of pyrolysis does not require inert gas throughput for collection of the product gases

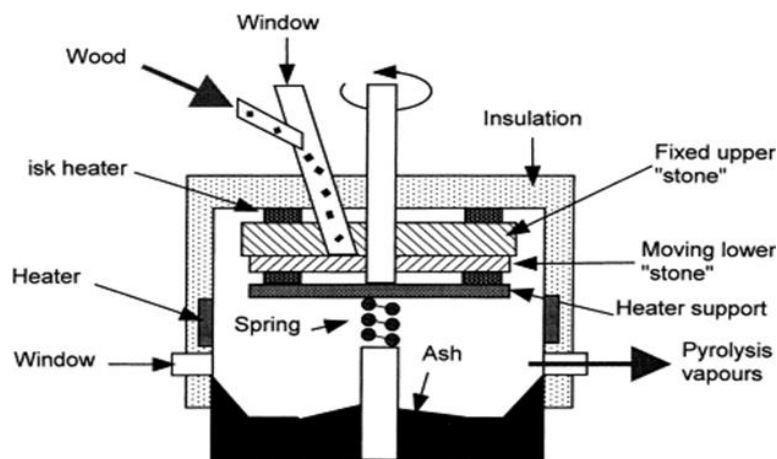


Figure-8: Ablative reactor.

–Vacuum Pyrolysis Reactor (Figure-9)

- represents a sluggish pyrolysis process with lower heat transfer rates conveyed with the fluidised bed technologies
- an induction and burner heater is used with molten salts
- the vapours formed are quickly detached from the vacuum
- it is categorized by lengthier residence time
- it is known to produce larger particles than most fast pyrolysis reactors
- there is also no requirement for carrier gas, and the process is mechanically complicated
- it needs high investment costs
- consistent operation of vacuum pyrolyser entails a superior feedstock input apparatus which discourages latent investors

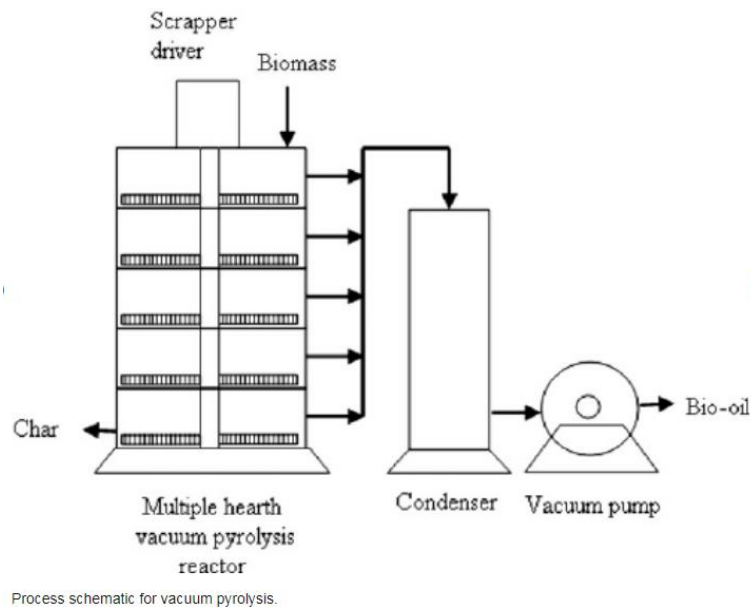


Figure-9: Vacuum Pyrolysis Reactor

–Rotating Cone Reactor

- it is a rotating cone reactor innovative reactor for flash pyrolysis with tiny char formation
- biomass feedstocks, such as rice husks, wood, palm kernel, coffee husk, and so on, can be milled in the rotating cone reactor
- there is no big scale of commercial implementation for a rotating cone reactor
- high-speed rotation provokes dynamic mixing of biomass that sequentially proceeds to fast heat transfer
- Figure-10 shows the rotating cone reactor

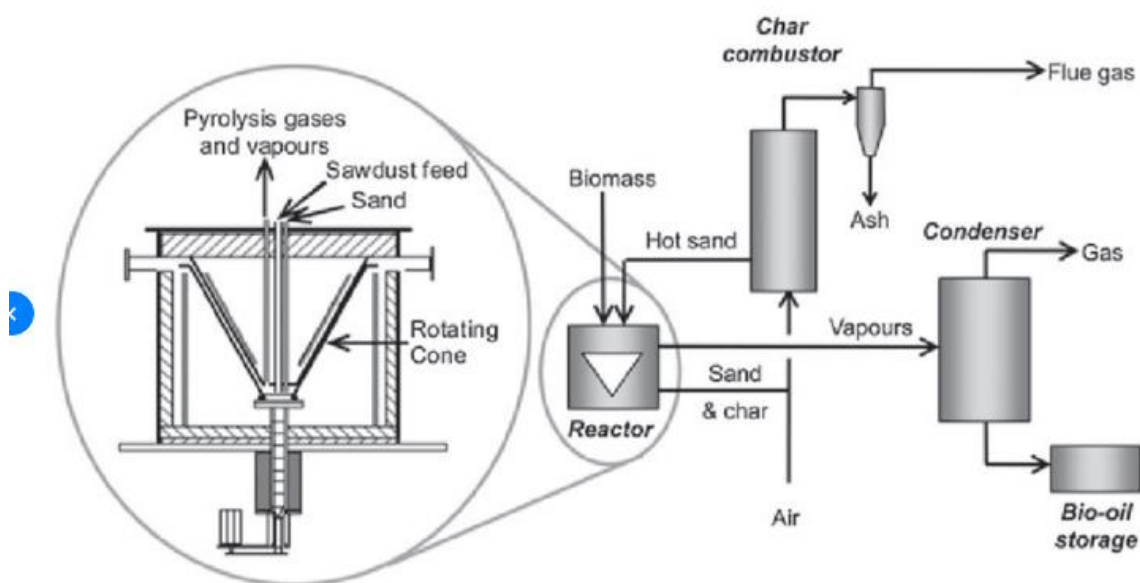
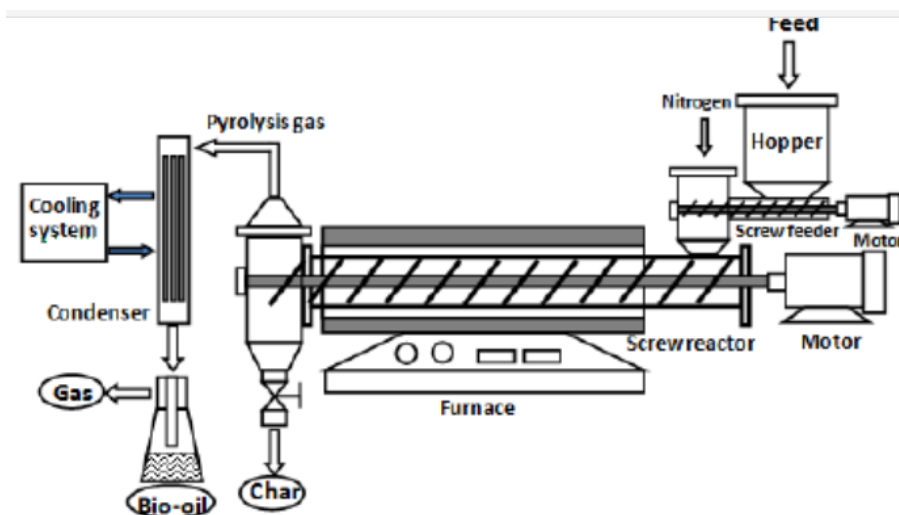


Figure-10: Rotating cone pyrolyser.

–Auger Reactor

- it is used to interchange biomass feedstock over an oxygen-free cylindrical tube
- in this pyrolyzer system, vapour residence time could be altered by fluctuating the heated zone
- auger reactors are getting attention of many mid-size industries
- stirring parts in the hot precinct and temperature transmission on a large scale are the major challenges for the auger reactors
- Figure-11 shows the auger reactor



– Comparison of various Pyrolyzers

The comparison is shown in the Table-6, below:

Table-6: Comparison of Various Biomass Pyrolysis Reactors Based on overall Performance and Efficiency

Pyrolyser	Status (units)	Bio-Oil Yield (wt %)	Operational Complexity	Particle Size	Biomass Variability	Scale-Up	The Inert Gas Flow Rate
Fixed bed	Pilot (single), lab (multiple)	75	Medium	Large	High	Hard	Low
Fluidised bed	Demo (multiple), lab (multiple)	75	Medium	Small	Low	Easy	High
Recirculating bed	Pilot (multiple), lab (multiple)	75	High	Medium	Low	Hard	High
Rotating cone	Demo (single)	70	Medium	Medium	High	Medium	Low
Ablative	Pilot (single), lab (multiple)	75	High	Large	High	Hard	Low
Screw / auger reactor	Pilot (multiple), lab (multiple)	70	Low	Medium	High	Easy	Low
Vacuum	Pilot (single), lab (few)	60	High	Large	Medium	Hard	Low

*Current Status of Pyrolysis Technology

- depleting condition of fossil fuels, food security and environmental concerns have posed great challenge to world community
- not only the quality, but the sustenance of life is under threat
- biomass pyrolysis will possibly help reduce CO₂ and the world's dependence on oil production
- bio-oils have the potential to lower CO₂ discharges; they are derived from plants which use CO₂ for growing
- an amalgamation of technologies is required to assimilate reactor design and operational procedure to recover the efficiency of biomass
- fast pyrolysis systems process small elements to maximize bio-oil yield, whereas low pyrolysis technologies use wood to produce char chunks
- the recognition of the environmental matters are allied with the use of carbonization technologies and the technical difficulties of operating fast pyrolysis reactors
- intermediate pyrolysis reactors propose prospects for the extensive balanced production of bio-oil and char
- presently, the foremost interests in pyrolysis technology are for CO₂ mitigation, electricity generation from biomass, and energy independence
- the pyrolysis technologies can be considered as slow, intermediate, fast, and flash pyrolysis but, then again, the most frequently used systems, meanwhile, are the fast and slow pyrolysis processes
- biochar is the key product of the slow pyrolysis, and transpires with moderate temperature, longer residence time, and small heating system rate
- dissimilarly, bio-oil is the key product of fast pyrolysis which formed with a fast heating rate within short residence time
- fast pyrolysis produces a higher quality and quantity of bio-oil than the slow pyrolysis
- it is expected that environmental and economic performance will increase the effectiveness of the pyrolysis process
- various actions are needed to overcome the technical challenges, including plummeting parasitic energy losses, improving pyrolysis reactor outlines, improving feedstock logistics, and enhancing biomass heating rate
- biomass feedstocks are most important to increase the pyrolysis products on a large scale
- this can be attained by producing energy-condensed products from biomass
- accumulation of metal and ash in reactor bed materials impedes pyrolysis which can reduce bio-oil yields
- controlling pyrolysis temperature and heating rate, and using smaller particle sizes can reduce accumulation
- recently, a study revealed that an ablative reactor can convert entire wood chips and produce more energy
- to conclude, cohesive pyrolysis systems that associate gasification or fast pyrolysis are one more important approach for making pyrolysis commercially viable and improving environmental performance
- Table-7 shows the available pyrolysis plants worldwide.

Table-7: Current Pyrolysis Plants Worldwide

Reactor Technology	Organisation/Location	Capacity (kg/h)	Desired Product
Fixed bed	Bio-alternative, USA	2000	Char
	THEE	500	Gas
	Dyna Motive, Canada	400	Oil
	BEST Energy, Australia	300	Oil
	Wellman, UK	250	Oil
Bubbling fluidised bed	Union Fenosa, Spain	200	Oil
	Zhejiang University, China	20	Oil
	RTI, Canada	20	Oil
	Waterloo University	3	Oil
	Zhejiang University, China	3	Oil
Circulating fluidised bed	Red Arrow, WI; Ensyn	1700	Chemicals
	Red Arrow, WI; Ensyn	1500	Chemicals
	Ensyn Engineering	30	Oil
	VTT, Finland, Ensyn	20	Oil
Rotating cone	BTG, Netherlands	200	Oil
	University Twente	10	Oil
Vacuum	Pyrovac, Canada	350	Oil
	Laval University	30	Oil
Ablative	PYTEC, Germany	250	Oil
	BBC, Canada	10–15	Char
	PYTEC, Germany	15	Oil
Vortex	Solar energy research Ins.	30	Oil
Another type	Fortum, Finland	350	Oil
	University Zaragoza	100	Gas
	Georgia Tech. Research Ins.	50	Oil

*Future Challenges

- in order to achieve full potential of biomass pyrolysis technology, improved understanding and successful commercialisation, additional research and development are needed.
- a couple of issues such as the lack of markets for pyrolysis oils and lack of biochar-derived products with well-defined performance characteristics must be overcome
- it is recommended to speed up the improvement and deployment of bio-oil refineries
- the improvement of flexible designs for pyrolysis units for producing higher yields of the product is a technical challenge
- as different pyrolysis technologies have different ranges of product yields, the selection of pyrolysis technologies, feedstocks, and their operating parameters should be based on the economic trade-offs
- however, in addition to the fundamental challenges, a few more important challenges for future biomass pyrolysis research are listed below:
 - understanding the proper working of pyrolysis reactors and processes
 - development of a new reactor that is cost-effective and highly efficient
 - development of catalysts for bio-oil upgrading
 - development of proper solar system reactors
 - post-pyrolysis processing to improve product bio-oil properties
 - understanding the limitations and potential for improvements of the quality of products obtained by biomass pyrolysis

- development of both fast pyrolysis and bio-oil upgrading, ensuring these are focused on delivering useful and valuable products

For the full implementation of pyrolysis technology, more research is needed to determine designs that will remove oxygen in the gas phase from pyrolysis oil. Pyrolysis technology has the potential to be applied in a vast diversity of situations and, through this process, diversity of products can be obtained. Hence, it is quite difficult to explore a sustainable design for all prospective applications. In addition, balanced financial investments to create new knowledge, technology, and markets for the purpose of building a united vision for the utilisation of pyrolysis technologies is crucial.